Predicting UFC Fights

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The Dataset

- **6,855 UFC fights**, with information about both fighters and the outcome of each fight.
- **Response Variable**
  - Fight_outcome_clean
  - 1 = Fighter 1 won
  - 0 = Fighter 1 lost
- **Explanatory Variables**
  - **11** numeric differential variables
  - Each calculated as Fighter 1 - Fighter 2
  - Categorical variable: Fighter1 stance

Positive Differential = Fighter 1 has the higher value; negative differential = Fighter 2 has the higher value



Methods

Why OLR is NOT Appropriate

- Y only takes on values of 0 and 1.
- That is incompatible with the normal distribution, which is continuous and symmetric.
- Since Y is Bernoulli, $Var(Y_i) = \pi_i * (1 - \pi_i)$
- The variance changes with π_i, while OLR assumes a constant variance σ^2
- OLR uses an identity link function. There is nothing stopping estimated outcomes being greater than 1 or less than 0.

Random Variable

Let Y_i be the random variable corresponding to whether Fighter 1 won fight *i*

won fight *i*.

$$Y_i = \begin{cases} 1 \text{ if Fighter1 won fight } i \\ 0 \text{ if Fighter2 won fight } i \end{cases}$$

Let $\pi_i = P(Y_i = 1)$

Where π_i represents the probability that Fighter 1 wins fight *i*

Why Logistic Regression is Appropriate

- Y_i only takes the values 0 and 1
- Therefore, Y_i has a Bernoulli distribution
- We use a Bernoulli random component
- We use a logit link function to connect the linear predictor to the probability that Fighter 1 wins
- The logit link keeps predicted probabilities between 0 and 1

Therefore: Bernoulli response + logit link = logistic regression

Link Function

$n_i = \log(\pi_i / (1 - \pi_i))$

In our model,

π_i = *P(*Fighter 1 wins fight *i*)

The logit link allows the linear predictor to take any real value while the predicted probability remains between 0 and 1.



Complete Separation: What We Needed to Rule Out

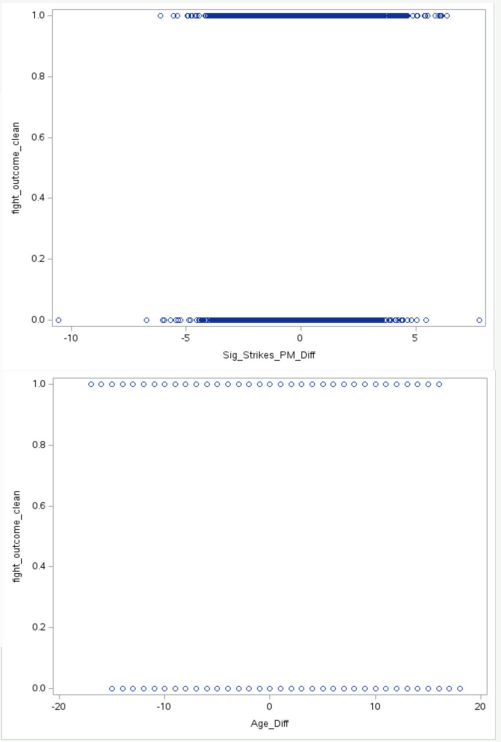
- Complete separation is when an explanatory variable, or a combination of predictors, perfectly separates fights where Fighter 1 lost (Y=0) from fights Where Fighter 1 Won (Y=1)
- Why it matters:
  - Standard maximum-likelihood estimates can become extreme and their variances /standard errors can inflate.
- If observed:
  - Firth's penalized likelihood.
- Test categorical and numeric predictors
  - Categorical: Table of fighter1_Stance * fight_outcome_clean => Looking for empty cells
  - Numeric:Plot of X variable * fight_outcome_clean =>Looking for one side containing only Y=0, or Y = 1

Pre-Model Diagnostics

- **No Complete Separation was observed**
- Categorical Variable Observation:
  - Every Stance level contains **Both** outcomes
- Numeric Variable Observation:
  - There is no single vertical cutoff that cl separates wins from losses
  - Both Y=0 and Y=1 appear across overlapping x values
  - **All 11 explanatory variables passed the visual separation check**

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of fighter1_Stance by fight_outcome_clean		
	fight_outcome_clean		
fighter1_Stance	0	1	Total
Open Stance	3 0.04 27.27 0.11	8 0.12 72.73 0.19	11 0.16
Orthodox	2007 29.28 39.48 75.79	3077 44.89 60.52 73.14	5084 74.16
Southpaw	487 7.10 35.06 18.39	902 13.16 64.94 21.44	1389 20.26
Switch	151 2.20 40.70 5.70	220 3.21 59.30 5.23	371 5.41
Total	2648 38.63	4207 61.37	6855 100.00



Our Logistic Regression Model

Random Component

$Y_i \sim \text{Bernoulli}(\pi_i)$



Systematic Component

$$g(\pi_i) = \underbrace{\log(\pi_i/(1-\pi_i))}_{\text{Link Function}} = \underbrace{\beta_0 + \beta_1\,\text{StanceOpen} + \beta_2\,\text{StanceOrthodox} + \beta_3\,\text{StanceSouthpaw} + \beta_4\,\text{Reach} + \beta_5\,\text{Sig_Strikes} + \beta_6\,\text{Strike_Accuracy} + \beta_7\,\text{Strikes_Absorbed} + \beta_8\,\text{Striking_Defense} + \beta_9\,\text{Takedown_Avg} + \beta_{10}\,\text{Takedown_Accuracy} + \beta_{11}\,\text{Takedown_Defense} + \beta_{12}\,\text{Submission_Avg} + \beta_{13}\,\text{Height} + \beta_{14}\,\text{Age}}_{\text{Linear Predictor}}$$

Switch is the baseline Stance. All numeric variables are Fighter 1 - Fighter 2 differentials.

Systematic Component Continued

- $g(\pi_i) = \log(\pi_i/(1-\pi_i)) = 0.2548 + 0.4005(\text{StanceOpen}) + 0.1728(\text{StanceOrthodox}) + 0.3866(\text{StanceSouthpaw}) + 0.0283(\text{Reach}) + 0.2893(\text{Sig_Strikes}) + 0.0154(\text{Strike_Accuracy}) - 0.2592(\text{Strikes_Absorbed}) + 0.00888(\text{Striking_Defense}) + 0.1102(\text{Takedown_Avg}) - 0.00114(\text{Takedown_Accuracy}) + 0.00825(\text{Takedown_Defense}) + 0.1119(\text{Submission_Avg}) - 0.0143(\text{Height}) - 0.0621(\text{Age})$

Link Function

Linear Predictor

Post-Model Diagnostics: No Complete Separation Observed

- Largest Standard Error observed was 0.6942 (Open Stance).
- 0.6942 < 5 by a large margin.
- Additionally, SAS's convergence criterion was satisfied
- **Conclusion: no complete separation is present in our model.**
- Continue using Standard Maximum-likelihood estimation, Firths isn't necessary

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
fighter1_Stance	Open Stance	1	0.4005	0.6942	0.3329	0.5640
fighter1_Stance	Orthodox	1	0.1728	0.1188	2.1163	0.1457
fighter1_Stance	Southpaw	1	0.3866	0.1295	8.9109	0.0028
Reach_Diff		1	0.0283	0.0106	7.0914	0.0077
Sig_Strikes_PM_Diff		1	0.2893	0.0241	144.1594	<.0001
Strike_Accuracy_Diff		1	0.0154	0.00321	22.9915	<.0001
Strikes_Absorbed_Per		1	-0.2592	0.0250	107.4451	<.0001
Striking_Defense_Dif		1	0.00888	0.00363	5.9645	0.0146
Takedown_Average_Dif		1	0.1102	0.0184	35.8555	<.0001
Takedown_Accuracy_Di		1	-0.00114	0.00117	0.9512	0.3294
Takedown_Defence_Dif		1	0.00825	0.00114	52.7658	<.0001
Submission_Average_D		1	0.1119	0.0336	11.1156	0.0009
Height_Diff		1	-0.0143	0.0139	1.0461	0.3064
Age_Diff		1	-0.0621	0.00545	129.8430	<.0001

Numeric Explanatory Variables

Sig_Strikes_PM_Diff = difference in significant strikes landed per minute

Strike_Accuracy_Diff = difference in striking accuracy

Strikes_Absorbed_Per_Minute_Diff = difference in strikes absorbed per minute

Striking_Defense_Diff = difference in striking defense

Takedown_Average_Diff = difference in takedown average

Takedown_Accuracy_Diff = difference in takedown accuracy

Takedown_Defence_Diff = difference in takedown defense

Submission_Average_Diff = difference in submission average

Reach_Diff = difference in reach (in inches)

Height_Diff = difference in height (in inches)

Age_Diff = difference in age (in years)



Categorical Explanatory Variable

- fighter1_Stance is a categorical explanatory variable with four levels:
  - Open stance
  - Orthodox
  - Southpaw
  - Switch
- SAS selected Switch as the baseline level.
- Open Stance, Orthodox, and Southpaw are interpreted relative to Switch

Results

Hypothesis Testing - Full Model

- Ho: $\beta_1 \ldots = \beta_{14} = 0$
- Ha: at least 1 of $\beta_1 \ldots \beta_{14} \neq 0$
- Test Statistic: lr = -2log(L($\beta_5$tilda) / L($\beta_5$hat)) = 918.07
- Null Distribution: $\chi^2(14)$
- P-Value < .0001
- Conclusion:
  - We reject Ho in favor of Ha as as the P-Value = .0001 < α.
  - Therefore, the full model has at least one explanatory variable related to the probability that fighter 1 wins.
  - Model is significant as a whole.

Testing Global Null Hypothesis: BETA=0			
Test	**Chi-Square**	**DF**	**Pr > ChiSq**
Likelihood Ratio	918.0695	14	<.0001
Score	838.6196	14	<.0001
Wald	732.1728	14	<.0001

For Coach: How to Interpret Logistic Regression Coefficients

- Odds Ratio = e^βj
- Holding all other factors constant, e^βj tells us how the odds of Fighter 1 winning change when the explanatory variable increases by 1 unit
- If OR > 1, the odds of Fighter 1 winning increase.
- If OR < 1, the odds of Fighter 1 winning decrease.



- Custom Odds Ratio = e^(c * βj)
- If OR > 1, the odds of Fighter 1 winning increase.
- If OR < 1, the odds of Fighter 1 winning decrease.
- For a c-unit change

Categorical Variable - Stances

- Switch vs Southpaw
- Odds Ratio: 1.472
- 95% Likelihood Confidence Interval for odds ratio: (1.142 to 1.897)
- What this means:
  - For an otherwise identical fighter, if fighter1 changes from the Switch stance to Southpaw, their odds of winning increase by a factor of 1.472 or 47%.

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
fighter1_Stance	Southpaw	1	0.3866	0.1295	8.9109	0.0028

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
fighter1_Stance Southpaw vs Switch	1.472	1.142	1.897

95% Likelihood Confidence Interval for odds ratios

$P(.132 \leq \beta_3 \leq .6399) = .95$

$P(e^{.132} \leq e^{\beta_3} \leq e^{.6399}) = .95$

$P(1.142 \leq e^{\beta_3} \leq 1.897) = .95$

Hypothesis Testing - Fighter Stance

- Categorical Variable
- Ho: $\beta_1 = \beta_2 = \beta_3 = 0$
- Ha: at least one of $\beta_1, \beta_2, \beta_3 \neq 0$
- Test Statistic: W = 13.5567
- Null Distribution: $\chi^2(3)$
- P-Value = 0.0036
- Conclusion:
  - We reject Ho in favor of Ha, as the p-value = .0.0036 < α = 0.05
  - So, fighter stance is a significant predictor of the fight outcome.

Numeric Variable - Significant Strikes Per Minute Difference

- Odds Ratio: 1.335
- 95% Likelihood Confidence Interval for odds ratio: (1.274 to 1.400)
- What this means
  - For an otherwise identical fighter, every 1 More Significant Strikes Landed Per Minute on their Opponent, their odds of victory Increases by 33.5%.
  - We are 95% confident that a one unit increase in significant strikes per minute is associated with the odds changing by a factor between (1.274 to 1.400).
  - For an otherwise identical fighter, every 2 More Significant Strikes Landed Per Minute on there Opponent, their odds of victory Increases by 78.4%.

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
Sig_Strikes_PM_Diff	1.335	1.274	1.400

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
Sig_Strikes_PM_Diff		1	0.2893	0.0241	144.1594	<.0001

Hypothesis Testing - Significant Strikes Landed Per Minute Difference

- Numeric Variable
- Ho: $\beta_5 = 0$
- Ha: $\beta_5 \neq 0$
- Test Statistic: W = 144.1594
- Null Distribution: $\chi^2(1)$
- P-Value: <.0001
- Conclusion:
  - We reject Ho in favor of Ha as the P-Value < α = .05.
  - So, Significant Strikes Landed Per Minute Difference is a significant predictor of the fight outcome.

Numeric Variable - Age Difference

- Odds ratio: 0.94
- 95% Likelihood Confidence Interval for odds ratio: (.93 to .95)
- What this means:
  - For an otherwise identical fighter, every 1 year older fighter1 is then fighter 2, their odds of victory decrease by 6%.
  - For an otherwise identical fighter, every 5 years older fighter1 is than fighter 2, their odds of victory decrease by about 26.7%.

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
Age_Diff		1	-0.0621	0.00545	129.8430	<.0001

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
Age_Diff	0.940	0.930	0.950

Custom Odds Ratio:

Odds at age x +5 / odds at age x = $e^{\beta_0+\ldots\beta_{14}(x+5)} / e^{\beta_0+\ldots\beta_{14}x} = e^{(5\beta_{14})} = e^{(5 \times -0.0621)} = 0.73308$

Custom Confidence Interval:

(e^5*-0.0728, e^5*-0.0515) =(0.6949, 0.7730) == (22.7%, 30.5%)

Numeric Variable - Reach Difference

- Odds Ratio: 1.029
- 95% Likelihood Confidence Interval for odds ratio: (1.008 to 1.050)
- What This Means?
  - For an otherwise identical fighter, every 1 Inch Longer Wingspan fighter1 has than fighter 2, their odds of victory Increase by 2.9%.
  - For an otherwise identical fighter, every 5 Inches Longer Wingspan fighter1 has than fighter 2, their odds of victory Increase by 15.2%.

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
Reach_Diff	1.029	1.008	1.050

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
Reach_Diff		1	0.0283	0.0106	7.0914	0.0077

e^(5*(0.0283) = e^0.1415 = OR 1.15200

Numeric Variable - Strikes Absorbed Per Minute Difference

- Odds Ratio: 0.772
- 95% Likelihood Confidence Interval for odds ratio: (.735 to .810)
- What This Means?
  - For an otherwise identical fighter, every Additional Strike Absorbed Per Minute by Fighter 1 relative to Fighter 2, decreases their odds of victory by 22.8%.

Odds Ratio Estimates

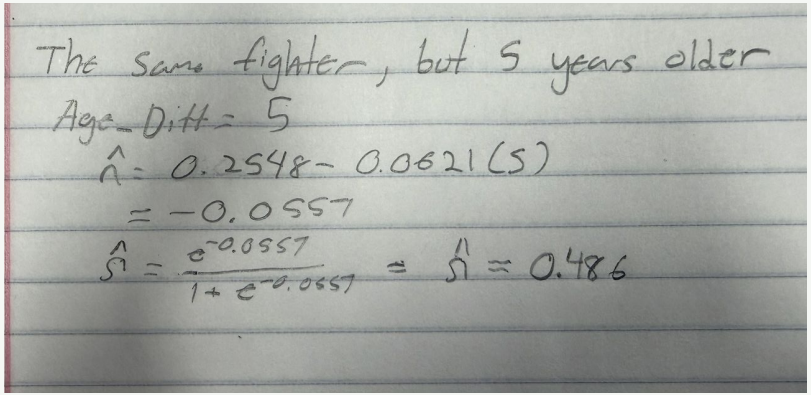
Effect	Point Estimate	95% Wald Confidence Limits	
Strikes_Absorbed_Per	0.772	0.735	0.810

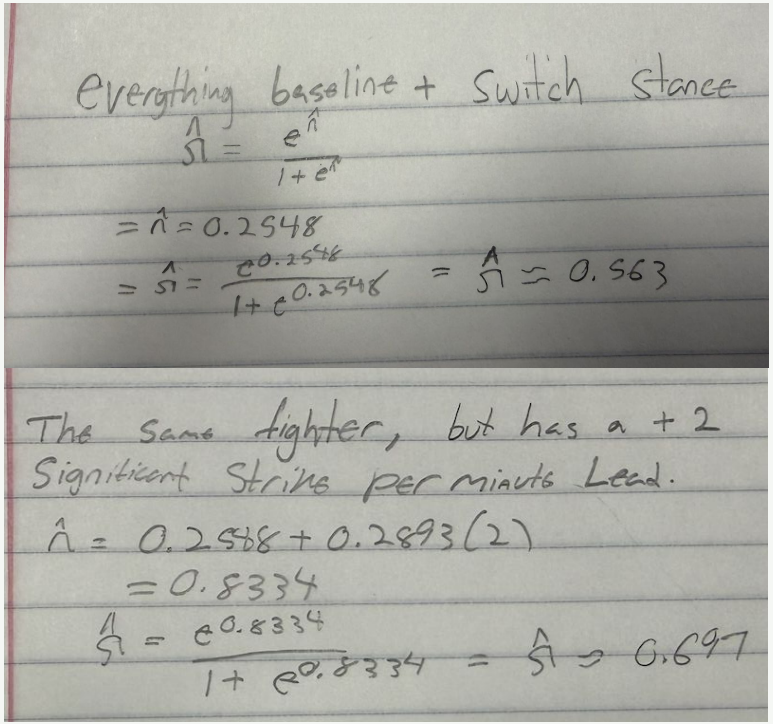
Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	0.2548	0.1145	4.9574	0.0260
Strikes_Absorbed_Per		1	-0.2592	0.0250	107.4451	<.0001

How Key Matchup Advantages Change Win Probability

- Otherwise-even matchup but switch stance: predicted Fighter 1 win probability = 56.3%
- Same Fighter… 5 years older: predicted win probability falls to 48.6%
- Same Fighter, but +2 significant strike lead per minute: predicted win probability rises to 69.7% (**an increase of 13.4 %)**

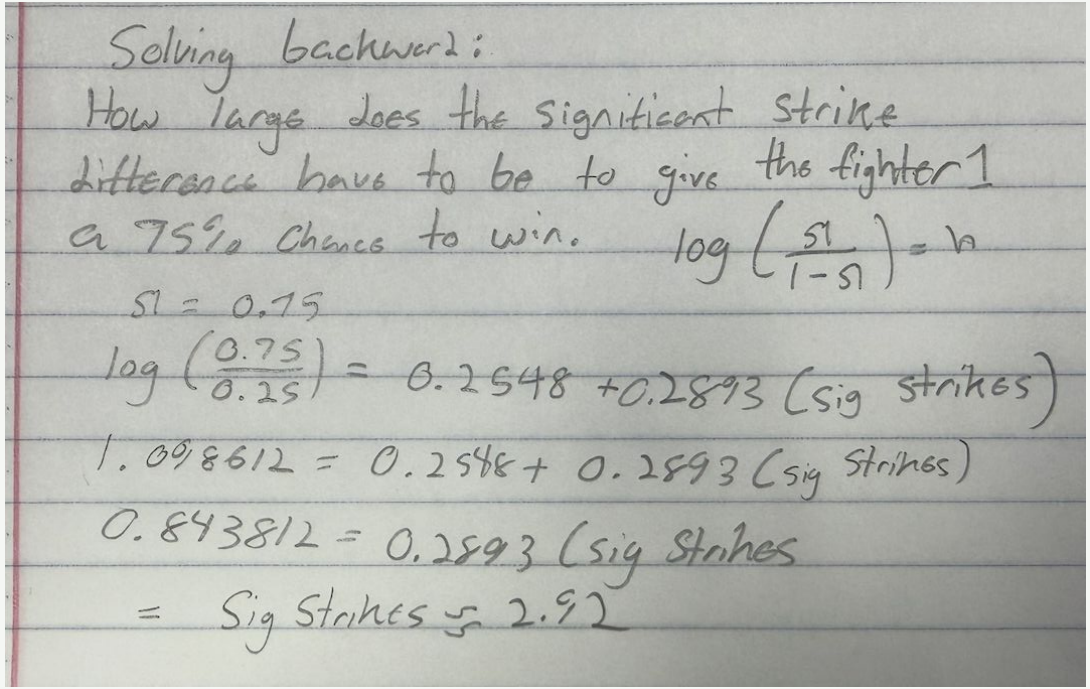




What Advantage Is Needed to Reach a Target Win Probability

Coach Takeaway:

- For an otherwise-even matchup, with Fighter 1 in a **Switch stance**, the model predicts that Fighter 1 needs about a **+2.92 significant strike per minute** advantage to reach a **75% predicted probability of winning**.



Dillon VS Nick

- $g(\pi_i) = \log(\pi_i/(1-\pi_i)) = = \beta_0 + \beta_1$ StanceOpen + β_2 StanceOrthodox + β_3 StanceSouthpaw + β_4 Reach + β_{13} Height + β_{14} Age
- We compared Dillon and Nick's differences to see how likely Dillon would be to beat Nick in a fight.
- .6209 = 0.2548 + 0.4005(1) + 0.0283(3) - 0.0143(4) - 0.0621(1)
- Converting to probability: = e^0.6209/ 1+ e^ 0.6209 = 0.6504
- We found based on Dillon's Stance, Height Advantage, Reach Advantage, and Age Disadvantage, that he has a 65.04% predicted probability to beat Nick in a fight holding all other values equal.



Conclusion: What Matters Most for Winning UFC Fights

- Significant striking advantage had one of the strongest relationships with winning fights.
  - For every 2 More Significant Strikes Landed Per Minute on Fighter 2 than Fighter 1, it increased Fighter 1's odds of victory by **78.4%.**
- Age Disadvantage lowered Fighter 1's changes
  - A 5 year disadvantage corresponded to about a **26.7%** decrease in the odds of winning.
- Reach advantage was beneficial
  - Each additional inch of reach advantage was associated with about 2.9% higher odds of winning.
- Fighter stance was significantly related to win probability
  - Southpaw Fighter 1's had about **47.2%** higher odds of winning than Switch Fighter 1's (holding all else constant).
- Absorbing more strikes hurt Fighter 1
  - Each additional strike absorbed per minute relative to Fighter 2 was associated with about **22.8%** lower odds of winning.
- Coach Takeaway: Use striking advantage, age, reach ,stance, and defensive performance together to evaluate how a matchup shifts a fighters predicted chance of winning.